

Decision Tree Algorithm

Agenda

- Decision Tree
- Splitting of Decision Tree
- Advantages and Disadvantages
- Homogeneity/Heterogeneity
- Calculation of Entropy and Gini
- Examples
- Hyperparameters used in Decision Trees
- Code walkthrough

Decision Tree

What is a Decision Tree?

Why Decision Tree?

Terminologies used in Decision Tree

- Root Node
- Leaf Node
- Branch

Types of Decision Trees

- Regression Tree
- Classification Tree

Types of Decision Trees

- Regression Tree
- Classification Tree

Summary

Splitting of Decision Trees

How does the decision tree split?

Example

Summary

Advantages and Disadvantages of Decision Trees

Advantages and Disadvantages of Decision Tree Algorithm

- Advantages

Advantages and Disadvantages of Decision Tree Algorithm

- Disadvantages

Summary

Homogeneity/Heterogeneity

What is Homogeneity in data?

Impurity

Metrics for Homogeneity

- Gini Index
- Entropy

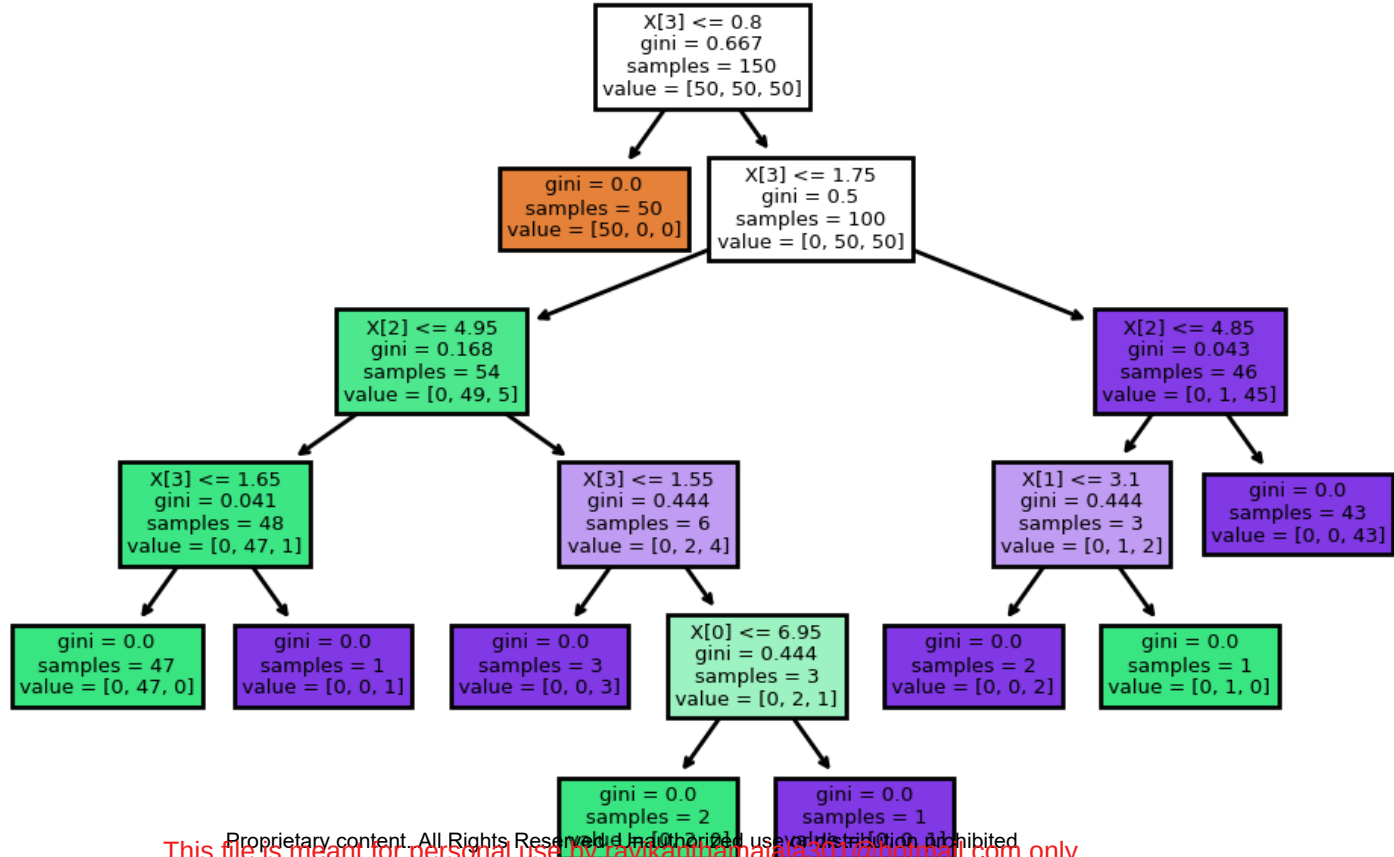
Gini Index

$$Gini = 1 - \sum_{i=1}^C (p_i)^2$$

Entropy

$$Entropy = \sum_{i=1}^C -p_i * \log_2(p_i)$$

Example



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Algorithms used

CART(Classification and Regression Tree)

C4.5

CHAID(Chi squared Automatic Interaction Detector)

Summary

Calculation of Gini and Entropy

Gini Index

$$Gini = 1 - \sum_{i=1}^C (p_i)^2$$

Entropy

$$Entropy = \sum_{i=1}^C -p_i * \log_2(p_i)$$

Entropy

Summary

Example

Example

Cust_ID	Gender	Occupati on	Age	Target
	1 M	Sal	22	1
	2 M	Sal	22	0
	3 M	Self-Emp	23	1
	4 M	Self-Emp	23	0
	5 M	Self-Emp	24	1
	6 M	Self-Emp	24	0
	7 F	Sal	25	1
	8 F	Sal	25	0
	9 F	Sal	26	0
	10 F	Self-Emp	26	0

Example

Example

Summary

Hyperparameters used in Decision Trees

Hyperparameters

- max_depth
- min_sample_split
- min_sample_leaf
- max_leaf_nodes

Summary

Pruning

What is Pruning?

Why Pruning?

Pre and Post Pruning?

How to perform Pruning?

Summary

Code Walkthrough

Summary

Thank you